Foundations of Programming

Assignment No.: 22

PROBLEM STATEMENT: Write a menu driven program in C to perform various string operations using in-built functions.

OBJECTIVE:

- To understand the concept of **strings in the C programming language**.
- To learn how to use **built-in string functions** for string manipulation.
- To perform operations such as **length calculation, copying, concatenation, comparison, and reversal** using menu options.

THEORY:

In the C programming language, a **string** is a sequence of characters terminated by a null character ('\0'). Strings are stored in **character arrays** and manipulated using functions available in the **string.h header file**.

Common **built-in string functions** include strlen() for finding the length of a string, strcpy() for copying one string to another, strcat() for concatenating two strings, and strcmp() for comparing two strings.

A **menu driven program** allows the user to select different operations from a list of options. Based on the user's choice, the corresponding string function is executed. This approach makes the program interactive and easy to use.

PLATFORM: Linux

ALGORITHM:

1. Start
2. Input a string str1
3. Repeat until user chooses Exit:
4. Display menu:
    1. Find length of string
    2. Copy string
    3. Concatenate strings
    4. Compare strings
    5. Reverse string
    6. Exit
5. Input user choice
6. Perform operation based on choice:
- If 1 → Use in-built function to find length and display it
- If 2 → Copy str1 into another string and display
- If 3 → Input another string str2, concatenate with str1, display result
- If 4 → Input another string str2, compare both strings, display result
- If 5 → Reverse the string and display
- If 6 → Exit loop
7. Stop

INPUT:

```
#include <stdio.h>
```

```c
#include <string.h>
int main() {
    char str1[100], str2[100];
    int choice;
    do {
        printf("\n--- String Operations Menu ---\n");
        printf("1. Compare Strings (strcmp)\n");
        printf("2. Concatenate Strings (strcat)\n");
        printf("3. Length of String (strlen)\n");
        printf("4. Copy String (strcpy)\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();
        if (choice >= 1 && choice <= 4) {
            printf("Enter string 1: ");
            scanf("%[^\n]", str1);
            getchar();
            if (choice == 1 || choice == 2) {
                printf("Enter string 2: ");
                scanf("%[^\n]", str2);
                getchar();
            }
        }
        switch (choice) {
            case 1:
                if (strcmp(str1, str2) == 0) {
                    printf("Result: Strings are equal\n");
                } else {
                    printf("Result: Strings are not equal\n");
                }
                break;
            case 2:
                strcat(str1, str2);
                printf("Result: Concatenated string: %s\n", str1);
                break;
            case 3:
                printf("Result: Length of string 1 is: %lu\n", strlen(str1));
                break;
            case 4:
                strcpy(str2, str1);
                printf("Result: String copied. New str2 is: %s\n", str2);
                break;
            case 5:
```

```
            printf("Exiting program.\n");
            break;
        default:
            printf("Invalid choice! Please try again.\n");
    }
  } while (choice != 5);
  return 0;
}
```

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100];
    int choice;

    do {
        printf("\n--- String Operations Menu ---\n");
        printf("1. Compare Strings (strcmp)\n");
        printf("2. Concatenate Strings (strcat)\n");
        printf("3. Length of String (strlen)\n");
        printf("4. Copy String (strcpy)\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();


        if (choice >= 1 && choice <= 4) {
            printf("Enter string 1: ");
            scanf("%[^\n]", str1);
            getchar();


            if (choice == 1 || choice == 2) {
                printf("Enter string 2: ");
                scanf("%[^\n]", str2);
                getchar();
            }
        }

        switch (choice) {
            case 1:

                if (strcmp(str1, str2) == 0) {
                    printf("Result: Strings are equal\n");
                } else {
                    printf("Result: Strings are not equal\n");
                }
                break;

            case 2:

                strcat(str1, str2);
                printf("Result: Concatenated string: %s\n", str1);
                break;

            case 3:

                printf("Result: Length of string 1 is: %lu\n", strlen(str1));
                break;

            case 4:

                strcpy(str2, str1);
                printf("Result: String copied. New str2 is: %s\n", str2);
                break;

            case 5:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 5);

    return 0;
```

OUTPUT: Enter first string: hello

Enter second string: world

String Operations Menu:

1. Length of string

2. Copy string

3. Concatenate strings

4. Compare strings

5. Reverse string

Enter your choice: 1

Length of first string = 5

Length of second string = 5

CONCLUSION:

Thus, the C program successfully demonstrates **different string operations using built-in functions** through a **menu driven approach**, making the program interactive and easy to use.

FAQs:

1. What are **built-in string functions** in C and why are they used?
2. What is the purpose of the **string.h header file** in C programs?
3. How does a **menu driven program improve user interaction** in programming?